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- ① b) data and pointer to the next node.
- ② c) NULL
- ③ b) self-referential structure
- ④ b) head == NULL
- ⑤ b) newnode → next = head; head = newnode;
- ⑥ b) temp = head; head = head → next; free(temp)
- ⑦ c) Traversing from tail to head.
- ⑧ b) n

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- ⑨ c) Copying the data of p → next into p, then deleting p → next.
- ⑩ b) while (p != NULL)
- ⑪ b) p == head
- ⑫
- ⑬ c) Three
- ⑭
- ⑮ b) p = (struct node*) malloc (size of (struct node));

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- (16) b) Avoid special-case handling for insertion and deletion at the head.
- (17) c) The first node
- (18) b) It does not support random (indexed) access.
- (19) b) Loss of access to all nodes, causing a memory leak.
- (20) b) A slow pointer and a fast pointer.
- (21) c) The last node.

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- (22) b) Nodes may be scattered anywhere in memory
- (23)

```
#include <stdio.h>
#include <stdlib.h>
struct node {
    int data;
    struct node * next;
};
```

(24) output :
30 → 50 → 70

~~(25)~~

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```
(25) void insertEnd (struct node * head, int x) {  
    struct node = t = (struct node) malloc (size of (struct  
        node));  
    t → data = x;  
    t → next = null;  
    if (* head == NULL) {  
        * head = t;  
        return;  
    }  
    struct node * p = * head  
    while (p → next != NULL)  
        p = p → next;  
    p → next = t;  
}
```

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```
(26) struct node * reverse  
    (struct node * head) {  
        struct node * prev = NULL;  
        struct node * curr = head;  
        struct node * next = NULL;  
        while (curr != NULL) {  
            next = curr → next;  
            curr → next = prev;  
            prev = curr;  
            curr = next;  
        }  
        return prev;  
}
```

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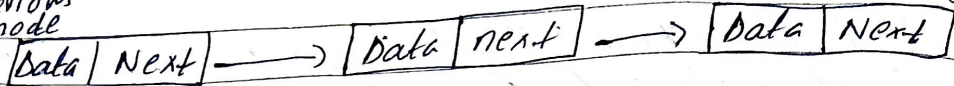
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(27)

Previous node

Next node



Pointer adjustment

$prev \rightarrow next = del \rightarrow next;$
 $free (del)$

(28)

$5 \rightarrow d \rightarrow 12 \rightarrow 3 \rightarrow NULL$

$7 \rightarrow 5 \rightarrow 8 \rightarrow 12 \rightarrow 3 \rightarrow NULL$

$7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 9 \rightarrow NULL$

Final $\rightarrow 7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 9 \rightarrow NULL$

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